

AMAR DHILLON

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Education

University of Maryland, College Park

B.S. in Computer Science, Minor in Statistics; GPA: 3.5

Expected May 2028

College Park, MD

Experience

Amazon Web Services

Sep 2026 – Dec 2026

Software Development Engineer Intern

Redmond, WA

- Building capacity planning for Amazon Redshift maintenance like patching: Java logic that sets upper and lower bounds on reserved EC2 capacity to balance availability against cost, validated by backtesting

Capital One

Jun 2026 – Aug 2026

Software Engineering Intern

Richmond, VA

- Cut AI workflow generation time **65%** to under 15s by replacing a Claude skill's open-ended search with focused sub-skills that call Python tools for workflow search and validation, guided by CloudWatch and OpenSearch latency logs
- Because the 270K+ row PostgreSQL workflow table had strict access limits, precomputed popular workflows into a category tree that the Claude skill searches first, with Redis caching full workflows to cut database queries
- Rebuilt the legacy tool developers use to create test-data workflows as a React and TypeScript Electron app with a visual React Flow editor and Jest-tested validation, and **90%** of 30+ surveyed developers preferred it

UMD App Development Contracting (Client: Amazon Leo)

Jan 2026 – Present

Software Engineer (Contract)

College Park, MD

- Cut end-to-end latency **34%** on a data pipeline that sends 6,000+ scraped research documents to model inference by streaming them through Amazon Kinesis into AWS Lambda instead of staging each step in S3
- Deployed a space-topic research tool for Amazon Leo's public-policy team and set it up for handoff with AWS CDK infrastructure-as-code and GitHub Actions CI/CD, so Leo engineers new to the project now maintain it on AWS Fargate
- Automated retraining for a production Amazon Bedrock classifier that rates research documents by importance: on an EventBridge schedule a larger LLM grades its ratings, and fine-tuning runs only when the grades shift past a threshold

UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

Machine Learning Research Intern

College Park, MD

- Cut research-environment setup time **30%** by pinning dependencies with uv and packaging the environment in Docker so every machine gets an identical setup

TerpLabs

Jun 2025 – Present

Co-Founder, Software Engineer

College Park, MD

- Built Redis caching in front of the Prisma ORM queries for Tortuga, the studio's class scheduler with 20,000+ visits in its first 30 days, to reduce database load and latency

Projects

IEX Order Book & Matching Engine | C++17, STL, POSIX, Linux perf

Jul 2026 – Aug 2026

- Built a C++17 order book and matching engine and replayed **50.9M** real IEX exchange messages through it without ever resyncing to the exchange's book, agreeing with the exchange on 99,122 of 99,123 fills and 100% of top-of-book prices
- Implemented B+ tree and LSM-tree backends for the event log and benchmarked them on identical 2M-record workloads, where the B+ tree gave **12.8x** the read throughput and 22x faster range queries at the cost of 27% more disk space
- Sustained **1.26M events/sec** end-to-end over the full replay, keeping each price level's orders in a std::list FIFO inside a std::map and a hash map from order ID to list iterator for O(1) cancels

Technical Skills

Languages: Java, Python, C++, TypeScript, JavaScript, SQL, Bash

Cloud & Infrastructure: AWS (EC2, Lambda, Kinesis, Fargate/ECS, API Gateway, EventBridge, SNS), observability (CloudWatch, X-Ray), AWS CDK (infrastructure as code), Docker, GitHub Actions (CI/CD), Linux

Backend, Full-Stack & Data: PostgreSQL, Redis, SQLAlchemy, Prisma, OpenSearch, REST APIs, Node.js, B+ tree and LSM-tree storage

Practices & Tools: debugging and performance profiling (perf, psutil), benchmarking, unit testing (Pytest, Jest), data structures and algorithms, Git, uv

Certification: AWS Certified Developer – Associate (Nov 2024 – Nov 2027)